

Emmanuel Asiimwe

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Profile

First-year Computational Engineering student at Aalto University, working toward numerical methods and scientific computing from a background in production backend engineering. What interests me is verification: knowing that a system does what it claims, whether that's a database enforcing its invariants or a solver converging to the right answer.

Education

Aalto University, BSc in Computational Engineering – Espoo, Finland 2026 – 2029 (expected)

Aga Khan High School, IB Diploma in Mathematics AA, Physics, Digital Society (HL) – Sep 2023 – Jul 2025
 Kampala, Uganda

- Score: 39/45
- Extended Essay: prime number distribution patterns in cryptography

Vienna College Namugongo, Cambridge IGCSE in 7 A*, 1 A, 1 B – Kira, Uganda Sep 2021 – Jun 2023

- Class of 2023 valedictorian
- Founded the school's first mathematics and coding clubs

Research interests

Numerical methods for partial differential equations · Computational fluid dynamics and atmospheric modelling · Verification and validation of scientific software · High-performance computing

Projects

PolyTrend Jun 2023 – May 2024

Polynomial regression and model-selection tool (Python, NumPy/SciPy, Qt GUI; GPL-3.0)

- Fits polynomial models by least squares and selects the degree by Bayesian Information Criterion, reporting R^2 , AIC/BIC and residual diagnostics
- Tested with pytest using deterministic, version-controlled fixtures; packaged with uv and PyInstaller
- Built for and distributed to a 50-student IB science cohort

Orurimi 2025 – present

Neural machine translation for Runyankore-Rukiga, a low-resource language (Python, PyTorch, SentencePiece)

- Building a parallel corpus and translation model for a language with little existing NLP tooling

Engineering experience

Co-founder & Backend Lead, Cadera Solutions Ltd (formerly Videra Digital) Jun 2025 – Jun 2026

Cadera, a school information system covering students, classes, grading and reporting; in production at one school since December 2025.

- Designed the PostgreSQL data model and NestJS service layer, including the business logic for grading, records and reporting
- Built the report-generation pipeline and record archiving

Skills

Scientific computing: NumPy, SciPy, Matplotlib, pytest, PyTorch

Languages: TypeScript, Python, JavaScript, C++

Backend: NestJS, Node.js, PostgreSQL, Prisma, REST API design

Tools: Linux, Git, Docker, uv